BOUNTIFUL

"Post a task. Pay the best answer."

The decentralized, guaranteed-payout bounty board operating on the BOT Blockchain. Pure onchain text storage, locked reward escrows, and automatic abandonment protection.

NETWORK	VERIFIED CONTRACT	PROTOCOL FEE
BOT Testnet (Chain 968)	0x9D67...Cab2	0.00% (100% to Workers)

Trustless Micro-Work for the BOT Ecosystem

Eliminating freelancer risk with game-theoretic onchain escrow and automatic abandonment guarantees.

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Irrevocable Escrow

Posters lock 100% of the native BOT reward in the smart contract. Posters can only cancel while submission count is exactly zero. Once any worker submits work, the funds cannot be retracted.

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Silent-Poster Protection

If a poster abandons a task and fails to pick a winner before the review deadline, anyone can trigger an automatic equal split of the reward across all contributors. Work is never left uncompensated.

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Zero Backend / Pure Onchain

No IPFS, no AWS, no database servers. Tasks, requirements, and solutions are stored directly in contract storage as plain text strings, ensuring permanent censorship resistance and zero downtime.

The Structural Flaws of Freelance Bounties

Freelancers and Web3 contributors face high risks and predatory commission models.

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The "Ghosting" Poster Trap

Freelancers frequently complete complex tasks (code, designs, translations) only for posters to disappear. In traditional escrow, funds sit frozen indefinitely or default back to the negligent poster.

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Excessive Intermediary Rake

Web2 platforms (Upwork, Fiverr) charge 10%–20% fees on both workers and clients. Even Web3 platforms impose 3%–5% protocol cuts, degrading micro-bounty feasibility.

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Offchain Fragility & Broken Links

Bounties that rely on centralized databases or unpinned IPFS hashes suffer from link rot, data loss, and API downtime, compromising historical proof of work.

Bountiful: Guaranteed Compensation

An autonomous, pull-based bounty protocol engineered for absolute fairness.

Poster Commitment Rules

- → **Upfront Reward Lock:** 100% of the prize is deposited upon creation.
- → **Anti-Rug Cancellation:** The poster can ONLY cancel when `submissionCount == 0`. Once any submission arrives, cancellation is permanently disabled.
- → **Flexible Review:** The poster can pick the winner anytime before the review deadline (even before submission close).

Worker Protection Architecture

- → **Automated Sunset Split:** If the review deadline passes without poster action, anyone can execute `splitIfAbandoned()`.
- → **Exact Reward Preservation:** Integer division remainders go to the 1st submitter; 0 wei is created or destroyed.
- → **Pull-Based Payouts:** Funds credit to `pendingBalance`. Reverting recipients cannot block others.

End-to-End Protocol Lifecycle

Deterministic state transitions governing tasks from deposit to settlement.

1

Create Task

Poster locks BOT, specifies submission window (1–90d) and review window (1–14d). Status = Open.

2

Submit Work

Workers submit text answers. 1 submission per wallet, max 50 per task. Poster can no longer cancel.

3

Review Window

Submissions close. Poster reviews all candidate solutions before the review deadline.

4

Winner Selection

Poster picks best solution. 100% of reward credited to winner's pendingBalance. Status = Awarded.

5

Abandon Split

If poster is silent after review deadline, anyone triggers equal split to all workers. Status = Split.

Defensive Smart Contract Engineering

Proven security patterns eliminating reentrancy, denial-of-service, and capital traps.

Pull-Payment Architecture

All payouts credit to `pendingBalance`. Funds leave exclusively when the user calls `withdraw()`. A worker contract that rejects BOT cannot block winner selection or split distributions.

Checks-Effects-Interactions

State balances are cleared and events emitted before any external ETH transfers. ReentrancyGuard is applied to all balance-altering functions, rendering recursive attacks mathematically impossible.

No Admin, No Upgrades

No owner, no admin keys, no pause functions, no upgrade proxies. Nobody—not even the protocol creators—can touch, redirect, or freeze user rewards.

Modern Full-Stack DApp Architecture

Engineered with Ethereum's most robust tooling and modern web standards.

Smart Contracts (Foundry)

- → **Language:** Solidity ^0.8.24 (optimized via solc).
- → **Libraries:** OpenZeppelin Contracts v5.0.2 (`ReentrancyGuard`).
- → **Testing:** Foundry test runner with 42 unit, fuzz, and stateful invariant suites.
- → **Deploy Script:** Automated Foundry broadcast script with keystore integration.

Frontend Client (Vite + React)

- → **Web3 Engine:** viem 2.21 (type-safe ABI contract interfaces).
- → **UI Framework:** React 18, Vite 5, TypeScript 5.
- → **Security UX:** Plain-text user input rendering (prevents XSS/HTML injection).
- → **Routing:** Hash-based deep routing (`#/bounty/:id`) for frictionless task sharing.

Synergy with the BOT Blockchain

Catalyzing onchain activity, token velocity, and decentralized developer coordination.

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Native BOT Velocity

Every bounty locks and transfers native BOT. As task volume expands, native BOT velocity and organic onchain demand increase across the network.

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Sub-Cent Micro-Bounties

BOT's high throughput and negligible gas fees make 0.05 BOT micro-tasks economically practical, unlocking a vibrant global gig economy unattainable on high-fee L1s.

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Community Coordination

An indispensable utility for BOT hackathons, bug bounties, logo contests, community translations, and technical writing without centralized coordinators.

Verification & Audit Readiness

Exhaustive testing matrix verifying mathematical invariants and onchain deployment.

TEST CATEGORY	SCENARIOS TESTED	RESULT
Unit & Boundary Validation	Title/Desc lengths (1–100, 1–1000B), day windows (1–90d, 1–14d), zero reward prevention, single submission per wallet, poster submission rejection.	**PASS (100%)**
State Transitions & No Double Settlement	Verified terminal states (Awarded, Cancelled, Split) strictly block all subsequent actions, submissions, or cancellations.	**PASS (100%)**
Split Remainder Math	Tested with 1, 2, 3, 7, and 50 submitters alongside odd/prime rewards; verifies 100% fund conservation with 0 wei loss.	**PASS (100%)**
Stateful Invariant Test (StdInvariant)	128,000 handler calls proving: `balance == sum(Open rewards) + sum(pendingBalance)` at all times.	**PASS (100%)**
BohrScan Contract Verification	Deployed on BOT Testnet (Chain ID 968) and verified via Blockscout API (GUID: 9d67ffa3...).	**VERIFIED (OK)**

Roadmap & Future Milestones

From testnet proof-of-concept to the foundational work coordination protocol of BOT Chain.

PHASE 1: COMPLETED

Testnet & Verification

- → Smart contract deployment on BOT Testnet (Chain 968).
- → Verification on BohrScan Blockscout explorer.
- → Vite SPA with hash routing, live counters, and wallet integration.

PHASE 2: UPCOMING

Mainnet & Ecosystem

- → Deploy Bountiful to BOT Mainnet (Chain ID 677).
- → Ecosystem hackathon & community task integration.
- → Community developer grants and liquidity bounties.

PHASE 3: EXPANSION

DAO & Reputation Layer

- → Onchain worker attestations and completion badges.
- → Multi-milestone task governance for DAOs.
- → Programmatic bounty funding SDK for BOT smart contracts.